

## MB 556: Natural Language Processing for Business

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| Course Code      | MB 556                                   |
| Course Name      | Natural Language Processing for Business |
| L-T-P-C          | 2-0-0-2                                  |
| Intended for     | MBA                                      |
| Mutual Exclusion | NA                                       |
| Approval         | 54th BoA (May 2024)                      |

### Course Contents

- **Fundamentals of Natural Language Processing:** Defining NLP and its tasks, history of NLP, Approaches to NLP, Understanding language, its syntax, structure and semantics, Language as data, Popular NLP applications. (4 Hours)
- **Text Representation, Preprocessing, and Transforming Models:** Information extraction and Text summarization, preprocessing: tokenization, Stemming, lemmatization, model building and evaluation, TF-IDF: Vectorizing, Bag of words, n-grams, Zipf's Law. (6 Hours)
- **Machines Understanding Words:** Embeddings: Text understanding, Representational Embeddings, Procedural Embeddings: Words to Vectors and Documents to Vectors, Textual Similarity. (6 Hours)
- **Deep Learning for NLP:** Sequential NLP and Episodic memory for NLP, Transformer Architecture, Transformer Encoder and Decoder, Attention mechanism, Transfer learning in NLP. (4 Hours)
- **Conversational AI:** Conversational AI Basics, Chatbots and Utterances, Taxonomy of chatbots, dialog and response generation. (6 hours)
- **NLP Applications:** Sentiment analysis, Content recommendations, NLP in healthcare, Supply chain, Law, Telecommunication, Education and Research. (4 Hours)

### Text Books

1. NA

### References

1. Jyotika Singh, **Natural Language Processing in the Real-World: Text Processing, Analytics, and Classification**, CRC Press, Chapman & Hall, 2023.
2. Ankur A. Patel and Ajay Uppili Arasanipalai, **Applied Natural Language Processing in the Enterprise: Teaching Machines to Read, Write & Understand**, O'Reilly Media, Inc., 2021.
3. Vajjala, S., Majumder, B., Gupta, A. & Surana, H., **Practical Natural Language Processing**, O'Reilly Media, Inc. 2020.
4. Masato Hagiwara, **Real-World Natural Language Processing: Practical applications with deep learning**, Manning Publications Co., 2022.
5. Dipanjan Sarkar, **Text Analytics with Python: A Practitioner's Guide to Natural Language Processing**, Apress, 2019.
6. Jacob Eisenstein, **Introduction to Natural Language Processing**, MIT Press, 2018.